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SURFACE AREA AND VOLUME OF SOLIDS

Regularly (10-12) questions have been asked from this chapter. Questions from this chapter are direct formula based, so one must be through with all the formulae and their application. It is easy to score section in the exam.



In this chapter, we will study how to find the surface area and volume of solid figures, like parallelepiped, cube, cuboid, cylinder, cone, frustum of cone, sphere and hemisphere and the combination of solid figures.

SOLID FIGURES

The objects which occupy space (i.e. they have three dimensions) are called solid figures. The solid figures can be derived from the plane figures.

A solid figure has surface area and volume.

Surface Area of Solid Figure

Surface area of a solid body is the area of all of its surfaces together. It is always measured in square units. Surface area is also referred to as the total surface area. The area of four walls or the area excluding top and bottom is called curved surface area or lateral surface area.

Volume of Solid Figure

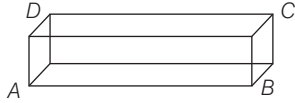
The measure of space occupied by a solid is called its volume. It is always measured in cubic units.

Cuboid

A figure which is surrounded by six rectangular surfaces, is called cuboid. It is also called rectangular parallelepiped.

- A cuboid has 8 corners, 12 edges, 6 faces and 4 diagonals.

- Volume of cuboid = Area of base $\times$ height = $l \times b \times h$
where, l = length, b = breadth and h = height



- Total surface area of cuboid = $2(lb + bh + lh)$
- Diagonal of the cuboid = $\sqrt{l^2 + b^2 + h^2}$
- Lateral surface area or Area of four walls = $2(l + b)h$.

EXAMPLE 1. The volume of a cuboid is 880 cm^3 , the area of its base is 88 cm^2 . Then, its height is

- a. 10 cm b. 12 cm c. 14 cm d. 16 cm

Sol. a. Here, volume of a cuboid = 880 cm^3

Area of the base = 88 cm^2

$$\text{Height} = \frac{\text{Volume of the cuboid}}{\text{Area of the base}} = \frac{880}{88} = 10 \text{ cm}$$

EXAMPLE 2. The length of the longest rod that can be placed in a room 12 m long, 9 m broad and 8 m high is

- a. 15 m b. 16 m
c. 17 m d. 17.5 m

Sol. c. Since, the length of the longest rod is the diagonal of the room

$$\begin{aligned} &= \sqrt{l^2 + b^2 + h^2} = \sqrt{(12)^2 + (9)^2 + (8)^2} \\ &= \sqrt{144 + 81 + 64} = \sqrt{289} = 17 \text{ m} \end{aligned}$$

EXAMPLE 3. The areas of three adjacent faces of a cuboid are x , y and z . If its volume is V , then which of the following is true?

- a. $V = x^3 y^2 z^2$ b. $V^2 = xyz$
c. $V = \sqrt[3]{xyz}$ d. $V = \frac{x^2 y}{z}$

Sol. b. Let the dimensions of a cuboid be l , b and h respectively, then volume of cuboid, $V = lbh$

Also, $x = lb$, $y = bh$ and $z = hl$

$$\therefore V = lbh$$

On squaring both sides, we get

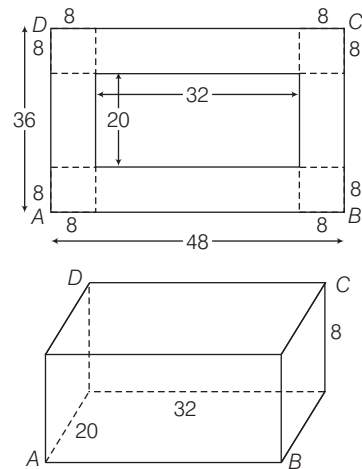
$$V^2 = l^2 b^2 h^2 = (lb) \cdot (bh) \cdot (hl) = xyz$$

EXAMPLE 4. A metallic sheet is of rectangular shape with dimensions $48 \text{ cm} \times 36 \text{ cm}$ from each of its corners a square of 8 cm is cut-off. An open box is made of the remaining sheet, what is the volume of the box?

- a. 13824 cm^3 b. 1728 cm^3
c. 5120 cm^3 d. None of these

Sol. c. Given length of metallic sheet = 48 cm

Breadth of metallic sheet = 36 cm



When square of side 8 cm is cut-off from each corner and the flaps turned up, we get an open box whose

$$\text{Length} = 48 - (8 + 8) = 32 \text{ cm}$$

$$\text{Breadth} = 36 - (8 + 8) = 20 \text{ cm}$$

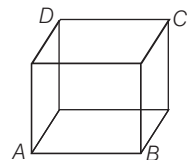
and height = 8 cm

$$\therefore \text{Volume of the box} = l \times b \times h = 32 \times 20 \times 8 = 5120 \text{ cm}^3$$

Cube

A cuboid whose length, breadth and height are same is called a cube.

- A cube has 6 surfaces, 12 edges, 8 corners and 4 diagonals.
- Cube is a special case of cuboid which has 6 square faces.
- Volume of a cube = $(\text{side})^3$
- Total surface area of a cube = $6 \times (\text{side})^2$
- Diagonal of a cube = $\sqrt{3} \times \text{edge}$
- Lateral surface area or area of four walls = $4 \times (\text{side})^2$



EXAMPLE 5. If the surface area of a cube is 96 cm^2 , then, its volume is

- a. 16 cm^3 b. 64 cm^3
c. 12 cm^3 d. 32 cm^3

Sol. b. Surface area of cube = $6 \times a^2$, where a = Side of a cube

$$\Rightarrow 6a^2 = 96 \Rightarrow a^2 = 16 \Rightarrow a = 4 \text{ cm}$$

$$\therefore \text{Volume of the cube} = a^3 = 4^3 = 64 \text{ cm}^3$$

EXAMPLE 6. The surface area and the length of the diagonal of the cube, if the volume of a cube is 2197 cm^3 , are

- a. 1012 cm^2 and 21.516 cm b. 1024 cm^2 and 24.516 cm
c. 1014 cm^2 and 22.516 cm d. None of these

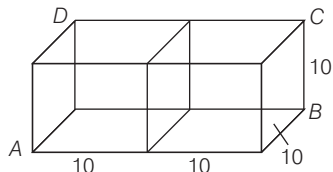
Sol. c. Volume of cube = (side)³ = 2197 cm³
 $\Rightarrow$ Side of cube = $\sqrt[3]{\text{volume of a cube}} = 13$ cm
 $\Rightarrow$ Surface area of cube
 $= 6 (\text{side})^2 = 6 (13)^2 = 6 \times 169 = 1014$ cm²
 and length of the diagonal of the cube = $\sqrt{3} \times \text{side}$
 $= \sqrt{3} \times 13 = 1.732 \times 13 = 22.516$ cm

EXAMPLE 7. How many 6 m cubes can be cut from a cuboid measuring 36 m $\times$ 15 m $\times$ 8 m?

- a. 10 b. 15
 c. 19 d. 20

Sol. d. Volume of given cuboid = $(36 \times 15 \times 8)$ m³
 Volume of the cube to be cut = $(6 \times 6 \times 6)$ m³
 $\therefore$ Number of cubes that can be cut from the cuboid
 $= \frac{\text{Volume of the cuboid}}{\text{Volume of the cube}} = \frac{36 \times 15 \times 8}{6 \times 6 \times 6} = 20$

EXAMPLE 8. Two cubes each of 10 cm edge are joined end-to-end. Then, the surface area of the resulting cuboid is



- a. 100 cm² b. 1000 cm²
 c. 2000 cm² d. None of these

Sol. b. Length of the resulting cuboid = $10 + 10 = 20$ cm
 Breadth of resulting cuboid = 10 cm
 Height of resulting cuboid = 10 cm
 $\therefore$ Surface area of resulting cuboid = $2(lb + bh + hl)$
 $= 2[20 \times 10 + 10 \times 10 + 20 \times 10]$
 $= 2[200 + 100 + 200] = 2 \times 500 = 1000$ cm²

IMPORTANT FORMULAE

- Volume of prism = Area of base $\times$ height
- Lateral surface area of prism = Perimeter of base $\times$ height
- Volume of pyramid = $1/3 \times$ Area of base $\times$ height
- Lateral surface area = $1/2 \times$ Perimeter of base $\times$ slant height

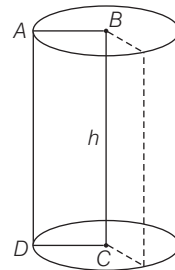
Right Circular Cylinder

A right circular cylinder is a solid figure obtained by revolving the rectangle, say $ABCD$ about its one side, say BC . Let base radius of right circular cylinder be r and its height be h . Then,

(i) Volume of the cylinder
 $= (\text{Area of base}) \times \text{Height}$
 $= \pi r^2 h$ cu units

(ii) Curved surface area = Circumference of the base $\times$ Height
 $= 2\pi r h$ sq units

(iii) Total surface area = Curved surface + Area of two ends
 $= 2\pi r h + 2\pi r^2 = 2\pi r (h + r)$ sq units



EXAMPLE 9. The volume of a cylinder is 448π cm³ and height 7 cm. Then, its lateral surface area and total surface area are

- a. 349 cm² and 753.286 cm²
 b. 352 cm² and 754.286 cm²
 c. 353 cm² and 755.286 cm²
 d. None of the above

Sol. b. Given, volume of the cylinder = 448π cm³
 and height of the cylinder = 6 cm

Let radius be r , then $\pi r^2 h = 448 \pi$

$$\therefore r^2 = \frac{448 \pi}{h \pi} = \frac{448}{7} = 64 \Rightarrow r = 8 \text{ cm}$$

$$\therefore \text{Lateral surface area of the cylinder} = 2\pi r h$$

$$= 2 \times \frac{22}{7} \times 8 \times 7 = 352 \text{ cm}^2$$

$$\text{Total surface area of the cylinder} = 2\pi r (h + r)$$

$$= 2 \times \frac{22}{7} \times 8 (7 + 8)$$

$$= 2 \times \frac{22}{7} \times 8 \times 15 = \frac{5280}{7} \approx 754.286 \text{ cm}^2$$

EXAMPLE 10. The radii of two cylinders are in the ratio 2 : 3 and their heights are in the ratio 5 : 3. Then, the ratio of their volumes is

- a. 4 : 9 b. 16 : 25 c. 20 : 27 d. None of these

Sol. c. For first cylinder,

Let radius = r_1 , height = h_1 , volume = V_1

For second cylinder,

Let radius = r_2 , height = h_2 and volume = V_2

$$\text{Then, } \frac{r_1}{r_2} = \frac{2}{3} \text{ and } \frac{h_1}{h_2} = \frac{5}{3} \Rightarrow r_1 = \frac{2r_2}{3} \text{ and } h_1 = \frac{5h_2}{3}$$

Required ratio of their volumes be $V_1 : V_2$.

$$\Rightarrow \pi r_1^2 h_1 : \pi r_2^2 h_2 \Rightarrow r_1^2 h_1 : r_2^2 h_2$$

$$\Rightarrow \frac{4}{9} r_2^2 \frac{5h_2}{3} : r_2^2 h_2 \Rightarrow \frac{20}{27} : 1 \Rightarrow 20 : 27$$

EXAMPLE 11. A cylindrical bucket of diameter 28 cm and height 12 cm is full of water. The water is emptied into a rectangular tub of length 66 cm and breadth 28 cm. The height to which water rises in the tub is

- a. 2 cm b. 4 cm c. 6 cm d. 8 cm

Sol. b. Volume of water in the bucket

$$= \pi r^2 h = \frac{22}{7} \times 14 \times 14 \times 12 = 7392 \text{ cm}^3$$

Let h be the height to which water rises in the tub.

$$\therefore \text{Volume of water in the tub} = 66 \times 28 \times h \text{ cm}^3$$

According to the question,

$$66 \times 28 \times h = 7392 \Rightarrow h = \frac{7392}{66 \times 28} = 4 \text{ cm}$$

Hence, the height to which water rises in the tub is 4 cm.

Hollow Cylinder

A hollow cylinder is one which is empty from inside and has some difference between the internal and external radius.

Let R and r be the external and internal radii of the hollow cylinder and h be its height.

Then,

$$(i) \text{ Volume of hollow cylinder} = \pi(R^2 - r^2)h \text{ cu units}$$

$$(ii) \text{ Curved surface area} \\ = 2\pi R h + 2\pi r h = 2\pi(R + r)h \text{ sq units}$$

$$(iii) \text{ Total surface area} \\ = 2\pi r h + 2\pi R h + 2\pi(R^2 - r^2) \text{ sq units} \\ = 2\pi(R + r)(h + R - r) \text{ sq units}$$

EXAMPLE 12. A hollow cylindrical tube, open at both ends is made of iron 1 cm thick. The volume of iron used in making the tube, if the external diameter is 12 cm and the length of tube is 70 cm, is

- a. 2420 cm³ b. 2520 cm³
c. 2720 cm³ d. 2900 cm³

Sol. a. Here, external radius (R_1) = 6 cm

and internal radius (R_2) = 6 - 1 = 5 cm

Height of hollow cylinder (h) = 70 cm

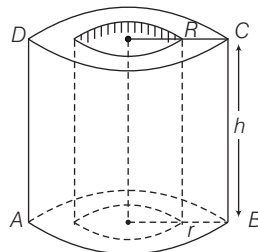
$\therefore$ Volume of iron used in making tube

= External volume - Internal volume

$$= \pi R^2 h - \pi r^2 h = \pi h(R^2 - r^2)$$

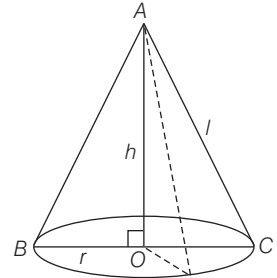
$$= \frac{22}{7} \times 70 \times (36 - 25)$$

$$= 220 \times 11 = 2420 \text{ cm}^3$$



Right Circular Cone

A right circular cone is a solid, generated by the revolution of a right angled triangle about one of its sides containing the right angle. Let height of a right circular cone be h , slant height be l and its radius be r . Then,



(i) The slant height of the cone,

$$l = AC = \sqrt{r^2 + h^2} \text{ units}$$

(ii) Volume of cone = $\frac{1}{3} \pi r^2 h$ cu units

(iii) Curved surface area of cone = $\pi r l$ sq units

(iv) Total surface area of a cone = Curved surface area + Base area = $\pi r(l + r)$ sq units

EXAMPLE 13. How many metres of cloth 5 m wide will be required to make a conical tent, the radius of whose base is 7 m and whose height is 24 m?

- a. 100 m b. 105 m c. 109 m d. 110 m

Sol. d. Given radius of base (r) = 7 m

and vertical height of tent (h) = 24 m

Slant height of the tent (l)

$$= \sqrt{r^2 + h^2} = \sqrt{24^2 + 7^2} = \sqrt{625} = 25 \text{ m}$$

$$\text{Now, curved surface area} = \pi r l = \frac{22}{7} \times 7 \times 25 = 550 \text{ m}^2$$

$\therefore$ Width of cloth = 5 m

$\therefore$ Length required to make conical tent

$$= \frac{550}{5} = 110 \text{ m}$$

EXAMPLE 14. The radius and the height of a right circular cone are in the ratio 5:12. If its volume is 314 m³, the slant height and the radius are

- a. 12 m, 5 m b. 13 m, 4 m c. 1 m, 4 m d. 13 m, 5 m

Sol. d. Let radius of cone be $5x$ and the height of the cone be $12x$.

$$\therefore \text{Volume of cone} = 314 = \frac{1}{3} \pi r^2 h$$

$$\Rightarrow \frac{1}{3} \times 3.14 \times (5x)^2 \times (12x) = 314 \Rightarrow 314x^3 = 314$$

$$\Rightarrow x^3 = 1 \Rightarrow x = 1$$

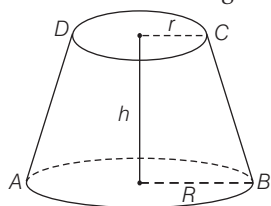
$\therefore$ Radius = 5 m and height = 12 m

$$\therefore \text{Slant height} = \sqrt{r^2 + h^2} = \sqrt{25 + 144} = \sqrt{169} = 13 \text{ m}$$

Hence, the slant height and the radius are 5 m and 13 m.

Frustum of a Cone

If a right circular cone is cut-off by a plane parallel to the base of the cone, then the portion of the cone between the cutting plane and base of the cone is called the frustum of the cone. Let R, r be the radii of base and top ($R > r$) of the frustum of a cone. Let height of frustum of a cone be h and slant height be l , then



- (i) Slant height of frustum of right circular cone

$$= BC = \sqrt{h^2 + (R-r)^2} \text{ units}$$

- (ii) Volume of frustum of right circular cone

$$= \frac{\pi h}{3} [R^2 + r^2 + Rr] \text{ cu units}$$

- (iii) Curved (lateral) surface area of frustum of right circular cone = $\pi l (R+r)$ sq units

- (iv) Total surface area of frustum of right circular cone
= Area of base + Area of top + Lateral surface area
= $\pi [R^2 + r^2 + l(R+r)]$ sq units

- (v) Total surface area of bucket = $\pi [(R+r)l + r^2]$ sq unit
[$\because$ it is open at the bigger end.]

EXAMPLE 15. The radii of the ends of a bucket of height 24 cm are 15 cm and 5 cm. Then, its capacity is

- a. 8000 cm³ b. 8100 cm³
c. 8171.43 cm³ d. 8200.43 cm³

Sol. c. Given, $h = 24$ cm, $R = 15$ cm and $r = 5$ cm

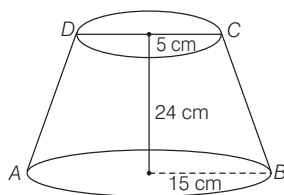
Capacity of bucket = Volume of frustum of a cone

$$= \frac{\pi h}{3} [R^2 + r^2 + Rr]$$

$$= \frac{22}{7} \times \frac{24}{3} [(15)^2 + 5^2 + 15 \times 5]$$

$$= \frac{22}{7} \times 8 [225 + 25 + 75]$$

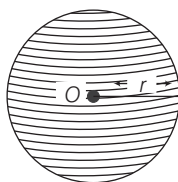
$$= \frac{176}{7} (325) = 8171.43 \text{ cm}^3$$



Sphere

The set of all points in space, which are equidistant from a fixed point, is called a sphere. Let radius of sphere be r , then

- (i) Volume of sphere = $\frac{4}{3} \pi r^3$ cu units



- (ii) Surface area of sphere = $4\pi r^2$ sq units

- (iii) Volume of a hollow sphere = $\frac{4}{3} \pi (R^3 - r^3)$ cu units

where, r = Inner radius and R = Outer radius

EXAMPLE 16. Given that the volume of a metal sphere is 38808 cm³. Then, its radius and its surface area are

- a. 7 cm and 616 cm² b. 21 cm and 5544 cm²
c. 14 cm and 2464 cm² d. None of these

Sol. b. Given, volume of the metal sphere = 38808 cm³

$$\text{But volume of sphere} = \frac{4}{3} \pi r^3$$

$$\therefore \frac{4}{3} \pi r^3 = 38808 \Rightarrow r^3 = 38808 \times \frac{3}{4} \times \frac{7}{22} = 9261$$

$$\Rightarrow r = (9261)^{1/3} = 21 \text{ cm}$$

$$\begin{aligned} \therefore \text{Surface area of the sphere} &= 4\pi r^2 \\ &= 4 \times \frac{22}{7} \times 21 \times 21 \\ &= 5544 \text{ cm}^2 \end{aligned}$$

EXAMPLE 17. How many balls each of radius 2 cm can be made by melting a big ball whose radius is 8 cm.

- a. 4 balls b. 16 balls c. 64 balls d. 128 balls

Sol. c. Given radius of big ball (R) = 8 cm

$$\therefore \text{Volume of big ball} = \frac{4}{3} \pi R^3 = \frac{4}{3} \pi (8)^3$$

$$\text{Radius of small ball } (r) = 2 \text{ cm}$$

$$\therefore \text{Volume of small balls} = \frac{4}{3} \pi (2)^3$$

$$\therefore \text{Required number of balls} = \frac{\text{Volume of big ball}}{\text{Volume of small ball}}$$

$$= \frac{\frac{4}{3} \pi (8)^3}{\frac{4}{3} \pi (2)^3} = \frac{8 \times 8 \times 8}{2 \times 2 \times 2} = 64 \text{ balls}$$

EXAMPLE 18. A copper sphere of diameter 18 cm is drawn into a wire of diameter 4 mm. Then, the length of the wire is

- a. 243 m b. 343 m c. 443 m d. 972 m

Sol. a. Let the length of wire be h cm.

$$\text{Volume of sphere} = \text{Volume of wire} \quad [\text{by condition}]$$

$$\Rightarrow \left(\frac{4}{3} \pi \times 9 \times 9 \times 9 \right) = \left(\pi \times \frac{2}{10} \times \frac{2}{10} \times h \right)$$

$$\begin{aligned} \Rightarrow \frac{h}{25} &= 972 \Rightarrow h = (972 \times 25) \text{ cm} \\ &= \frac{972 \times 25}{100} = 243 \text{ m} \end{aligned}$$

Hence, the length of the wire is 243 m.